君正®

ZRT 平台多 sensor 功能使用文档

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北京君正集成电路股份有限公司
Ingenic Semiconductor Co., Ltd.

君正®
SDK 开发指南
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Release history

Date	Revision	Change
Apr.2024	1.0	First release

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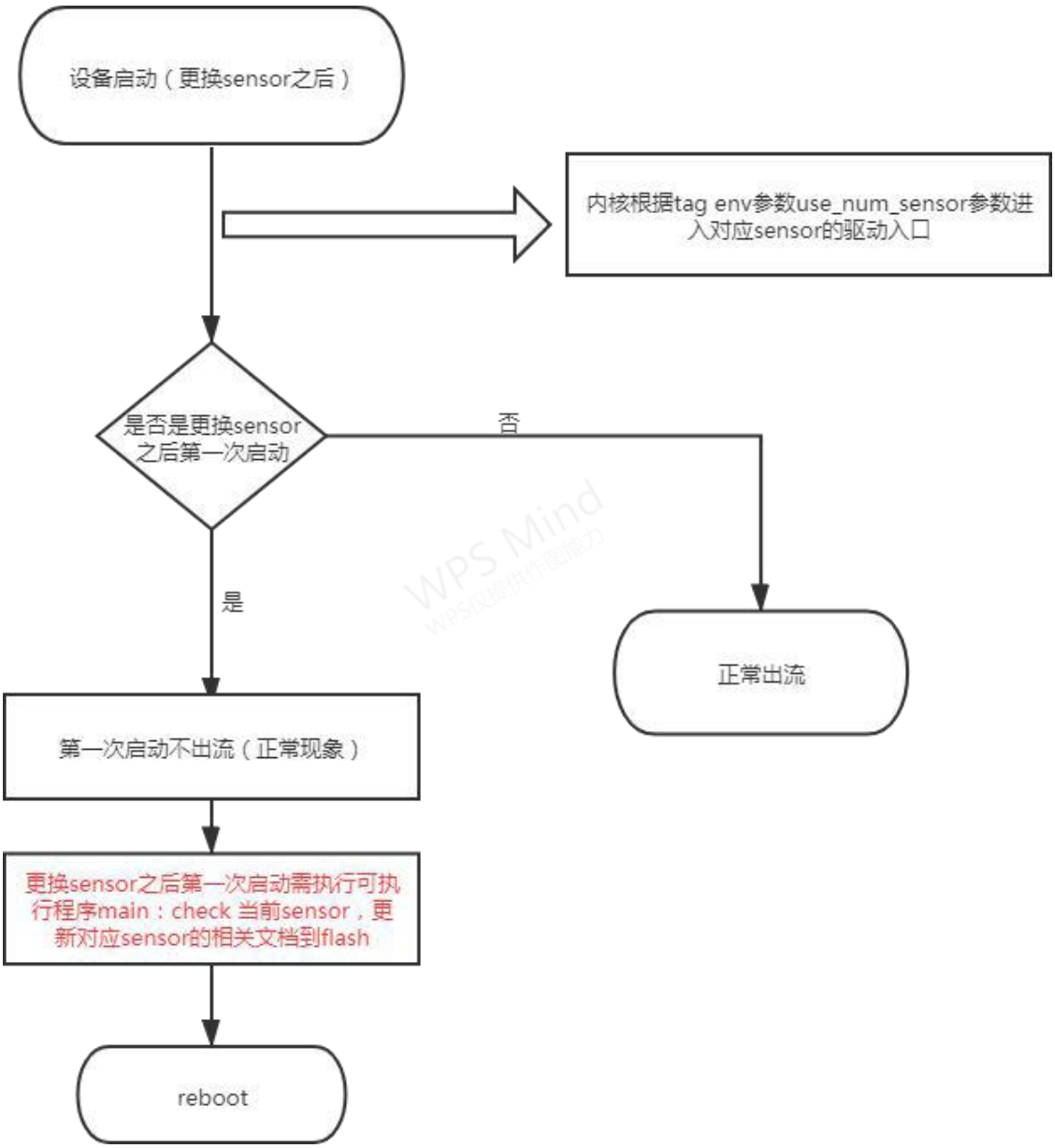
流程图：



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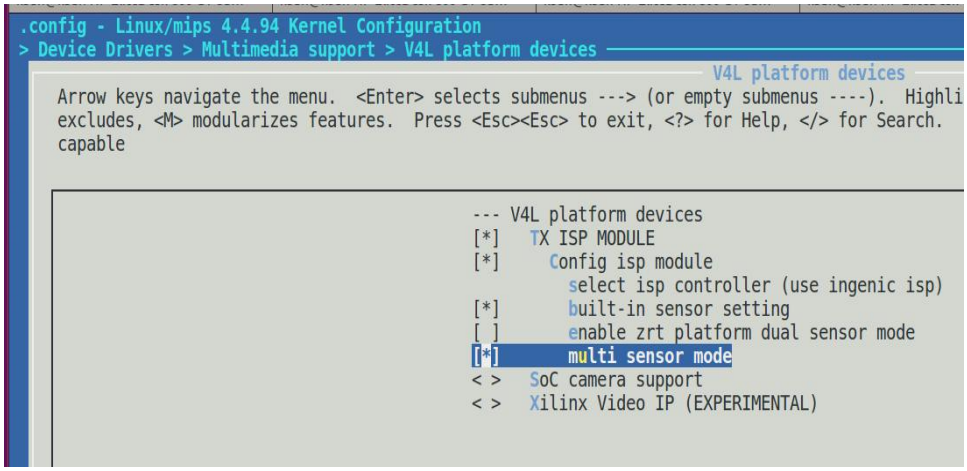
1.1 kernel 修改

1. 使用内核默认配置基础上选中 multi sensor mode 选项



注意：编译内核前需要先 source sensor.sh 脚本。

1.2 driver 修改

1：sensor 驱动修改

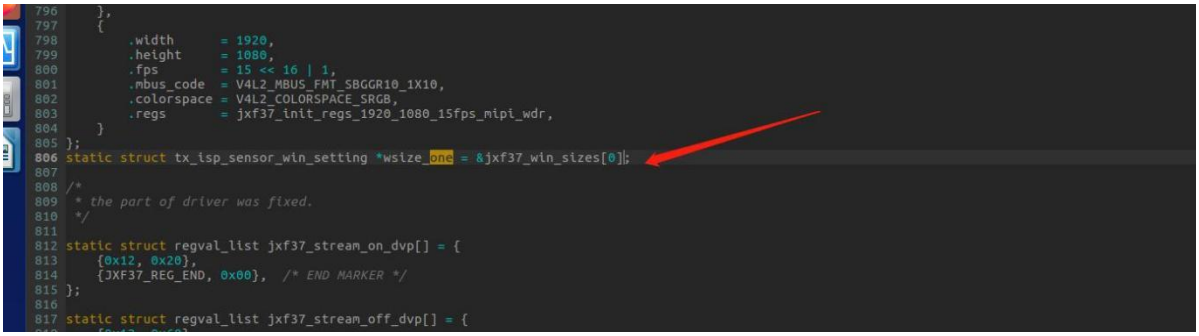
该文件提供几款已修改的兼容版本的 sensor 驱动，目录：os/drivers/sensor-t41/zeratul/multi_sensor 建议客户拿到资料之后自行跟原 sdk 的相同 sensor 驱动做对比。

如下图使用 jxf37 为例，该 sensor 的 demo 项目中第一款 sensor，即将相关接口和变量加上专属标识符_one，类比推理，第二款加上_two... 。用于区分不同 sensor 的驱动接口。

文档提供 jxf37（1）、gc2063（2）、sc3235（3）。三款已修改的 sensor 驱动，作为参考，根据项目需求，参考格式自行修改所需的 sensor 驱动。

修改完成之后各个 sensor 驱动重新编译。

make && make release



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```
    .video = &jxf37_video_ops,
    .sensor = &jxf37_sensor_ops,
};

/* It's the sensor device */
static u64 tx_isp_module_dma_mask = ~(u64)0;
struct platform_device sensor_platform_device_one = {
    .name = "jxf37",
    .id = -1,
    .dev = {
        .dma_mask = &tx_isp_module_dma_mask,
        .coherent_dma_mask = 0xffffffff,
        .platform_data = NULL,
    },
    .num_resources = 0,
};

static int jxf37_probe(struct i2c_client *client, const struct i2c_device_id *id)
{
    struct tx_isp_subdev *sd;
    struct tx_isp_video_in *video;
    struct tx_isp_sensor *sensor;
    int ret;

    sensor = (struct tx_isp_sensor *)kzalloc(sizeof(*sensor), GFP_KERNEL);
```

```
char * get_sensor_name_one(void)
{
    return "jxf37";
}

int get_sensor_i2c_addr_one(void)
{
    return 0x40;
}

int get_sensor_width_one()
{
    return 1920;
}

int get_sensor_height_one()
{
    return 1080;
}

extern struct list_head sensor_list;
extern struct add_sensor *sensors[8];

int init_sensor_one(void)
{
//  printk("======> enter init_sensor_one \n");
    int ret = 0;
    ret = private_driver_get_interface();
    if(ret){
        printk("Failed to init jxf37 dirver.\n");
        return -1;
    }

    return private_i2c_add_driver(&jxf37_driver);
}

struct add_sensor sensor_one = {
    .name = "jxf37",
    .sensor_init = init_sensor_one,
    .add_sensor_get_name = get_sensor_name_one,
    .add_sensor_get_i2c_addr = get_sensor_i2c_addr_one,
};

extern int use_num_sensor;

int sensor_register_one(void)
{
//  printk("====== enter sensor_register_one use_num_sensor = %d ======\n", use_num_sensor);

    if(data_type == TX_SENSOR_DATA_TYPE_WDR_DOL) {
        sensor_one.wdr_mode = 1;
    } else {
        sensor_one.wdr_mode = 0;
    }
~/work/zeratul_t31/os/drivers_num_sensor/sensor-t31/zeratul/jxf37/jxf37.c[POS=1605,5][98%]16/06/21 - 15:20
```

修改函数名，用于区分sensor driver的接口

2：isp 驱动使用

非直通使用：libtxisp-ms-t23.hex

直通　使用：libtxisp-t23-ms-direct.hex

如果没有该文件，需要找对应工程师基于上面kernel配置提供(ingenic_zeratul_t23_v2.0.0_20240415.7z版本以及之前没有ms命名)

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3：common.c 路径：os/drivers/sensor-t23/zeratul/common

```
11 #include <linux/delay.h>
12 #include <linux/syscalls.h>
13 #include <linux/fs.h>
14
15 #include "../include/tx-isp-common.h"
16 int sensor_register_one(void);
17 int sensor_register_two(void);
18 int sensor_register_third(void);
19 struct list_head sensor_list;
20 struct add_sensor *sensors[8];
21
22 extern int use_num_sensor;
23
24 int add_sensor_chose(void)
25 {
26 //  printk("======= enter add_sensor_chose common  use_num_sensor = %d ======= \n", use_num_sensor);
27     switch(use_num_sensor) {
28         case 0:
29             sensor_register_one();
30             break;
31
32         case 1:
33             sensor_register_two();
34             break;
35
36         case 2:
37             sensor_register_third();
38             break;
39
40         default:
41             printk("====== swicth no match ====== \n");
42             break;
43
44     }
45
46     struct add_sensor *use_sensor;
47     use_sensor = sensors[use_num_sensor];
48     use_sensor->sensor_init();
49
50     return 0;
51 }
52
53
54
55
56
~
~
~/work/zeratul_t31/os/drivers_num_sensor/sensor-t31/zeratul/common/common.c[POS=14,1][25%]16/06/21 - 20:44
"common/common.c" 56L, 1049C 已写入
```

该接口最后被isp调用

根据识别到的sensor号，执行对应的sensor register

根据实际项目需求自行添加接口，以及对应sensor 驱动里的实现函数

执行：make && make release

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1.3 tag 修改

1：变量介绍

use_multi_sensor: 该值与内核使用的 sensor 序号一一对应，内核获取到 env 参数 use_multi_sensor 进行判断，进入对应的 sensor 驱动入口函数。

2：制作tag分区的脚本为 由build_tag_t23.sh执行

该文档提供了制作 tag 的源代码。

```
if [ ${ZRT_ENV_BUILD_TYPE} = release ]; then
    QUIET=quiet
fi

if [ ${ZRT_ENV_BOARD_VERSION} = V2 ]; then
    UART=ttyS0
else
    UART=ttyS1
fi

if [ ${ZRT_ENV_SOC_TYPE} = SOC_T31Z ]; then
    CMDLINE="console=${UART},115200n8 mem=80M@0x0 rmem=48M@0x5000000 root=/dev/ram0 rw rdinit=/linuxrc mtdparts=jz_sfc:256K(boot),640K(tag),5M(kernel),3M(rootfs),2560K(recovery),4224K(system),
512k(config),16M@0(all) lpj=6955008 "
    BOOTINFO="kernel=5M@0xe0000 rootfs=3M@0x5e0000"
    ENVINFO="senv;[HW];init_vw=1920;init_vh=1080;nrvbs=3;mode=0;use_num_sensor=0;debug;eenv;"
    SOC_TYPE=t31

fi

if [ ${ZRT_ENV_SOC_TYPE} = SOC_T31A ]; then
    CMDLINE="console=${UART},115200n8 mem=80M@0x0 rmem=48M@0x5000000 root=/dev/ram0 rw rdinit=/linuxrc mtdparts=jz_sfc:256K(boot),352K(tag),5M(kernel),3M(rootfs),2560K(recovery),4512K(system),
512k(config),16M@0(all) lpj=6955008 ${QUIET}"
    BOOTINFO="kernel=5M@0x98000 rootfs=3M@0x590000"
    ENVINFO="senv;[HW];init_vw=1920;init_vh=1080;nrvbs=3;mode=0;eenv;"
    SOC_TYPE=t31
    RISCV_FW="../riscv_fw/riscv_fw_${ZRT_ENV_SENSOR}.bin"
fi

RISCV_FW="../riscv_fw/riscv_fw_num0.bin"
RISCV_FW_NUM1="../riscv_fw/riscv_fw_num1.bin"
RISCV_FW_NUM2="../riscv_fw/riscv_fw_num2.bin"

SENSOR_SETTING_MODE0="../sensor_calibration_bin/sensor_cal_num0.bin"
SENSOR_SETTING_MODE1="../sensor_calibration_bin/sensor_cal_num1.bin"
SENSOR_SETTING_MODE2="../sensor_calibration_bin/sensor_cal_num2.bin"

SENSOR_INIT_NUM0="../sensor_init_setting/sensor_init_num0.bin"
SENSOR_INIT_NUM1="../sensor_init_setting/sensor_init_num1.bin"
SENSOR_INIT_NUM2="../sensor_init_setting/sensor_init_num2.bin"

SENSOR_AE_TABLE="../sensor_start_ae_table/ae_table_num0.bin"
SENSOR_AE_TABLE_NUM1="../sensor_start_ae_table/ae_table_num1.bin"
SENSOR_AE_TABLE_NUM2="../sensor_start_ae_table/ae_table_num2.bin"

cd ${ZRT_ENV_TOOLS_DIR}/make_tag/riscv_fw/
make clean && make

cd ${ZRT_ENV_TOOLS_DIR}/make_tag/sensor_calibration_bin/
make clean && make
```

上述有文字说明

riscv固件

效果文件

init文件

ae_table表格

```
SENSOR_SETTING_MODE0="../sensor_calibration_bin/sensor_cal_num0.bin"
SENSOR_SETTING_MODE1="../sensor_calibration_bin/sensor_cal_num1.bin"
SENSOR_SETTING_MODE2="../sensor_calibration_bin/sensor_cal_num2.bin"

SENSOR_INIT_NUM0="../sensor_init_setting/sensor_init_num0.bin"
SENSOR_INIT_NUM1="../sensor_init_setting/sensor_init_num1.bin"
SENSOR_INIT_NUM2="../sensor_init_setting/sensor_init_num2.bin"

#SENSOR_AE_TABLE="../sensor_start_ae_table/ae_table_${ZRT_ENV_SENSOR}.bin"
SENSOR_AE_TABLE="../sensor_start_ae_table/ae_table_num0.bin"
SENSOR_AE_TABLE_NUM1="../sensor_start_ae_table/ae_table_num1.bin"
SENSOR_AE_TABLE_NUM2="../sensor_start_ae_table/ae_table_num2.bin"

cd ${ZRT_ENV_TOOLS_DIR}/make_tag/riscv_fw/
make clean && make

cd ${ZRT_ENV_TOOLS_DIR}/make_tag/sensor_calibration_bin/
make clean && make

cd ${ZRT_ENV_TOOLS_DIR}/make_tag/sensor_start_ae_table/
make clean && make

cd ${ZRT_ENV_TOOLS_DIR}/make_tag/sensor_init_setting/
make clean && make
#make ${ZRT_ENV_SENSOR}_${SOC_TYPE}_init
cp *.bin ${ZRT_ENV_TOOLS_DIR}/make_tag/tag_generator/

cd ${ZRT_ENV_TOOLS_DIR}/make_tag/tag_generator/
make distclean && make tag_generator_t31_host

if [ -f ${RISCV_FW} ]; then
    ./tag_generator_t31_host --create --tag_path=tag.bin --cmdline="${CMDLINE}" --bootinfo="${BOOTINFO}" --env="${ENVINFO}" --sensor_init=${SENSOR_INIT_NUM0} --sensor_init_num1=${SENSOR_INIT_NUM1} --
sensor_init_num2=${SENSOR_INIT_NUM2} --fwinfo="${ZRT_ENV_FW_VERSION}" --riscv_fw="${RISCV_FW}" --riscv_fw_num1="${RISCV_FW_NUM1}" --riscv_fw_num2="${RISCV_FW_NUM2}" --
sensor_setting_mode0="${SENSOR_SETTING_MODE0}" --sensor_setting_mode1="${SENSOR_SETTING_MODE1}" --sensor_setting_mode2="${SENSOR_SETTING_MODE2}" --aetab="${SENSOR_AE_TABLE}" --
aetab_num1="${SENSOR_AE_TABLE_NUM1}" --aetab_num2="${SENSOR_AE_TABLE_NUM2}"
else
    ./tag_generator_t31_host --create --tag_path=tag.bin --cmdline="${CMDLINE}" --bootinfo="${BOOTINFO}" --env="${ENVINFO}" --sensor_init=${ZRT_ENV_SENSOR}_${SOC_TYPE}_init.bin --
fwinfo="${ZRT_ENV_FW_VERSION}" --sensor_setting_mode0="${SENSOR_SETTING_MODE0}" --aetab="${SENSOR_AE_TABLE}"
fi

cp tag.bin tag_bk.bin
#dd if=./tag.bin of=./tag_cmp.bin bs=1K skip=384
#lzop -9 -f tag_cmp.bin -o tag_cmp.bin.lzo
#dd if=./tag_cmp.bin.lzo of=./tag.bin bs=1K seek=384

dd if=./tag.bin of=./tag_cmp1.bin bs=1K skip=384 count=288
dd if=./tag.bin of=./tag_cmp2.bin bs=1K skip=672 count=288
dd if=./tag.bin of=./tag_cmp3.bin bs=1K skip=960 count=288

lzop -9 -f tag_cmp1.bin -o tag_cmp1.bin.lzo
lzop -9 -f tag_cmp2.bin -o tag_cmp2.bin.lzo
lzop -9 -f tag_cmp3.bin -o tag_cmp3.bin.lzo

dd if=./tag_cmp1.bin.lzo of=./tag.bin bs=1K seek=384 count=80 conv=notrunc
dd if=./tag_cmp2.bin.lzo of=./tag.bin bs=1K seek=464 count=80 conv=notrunc
dd if=./tag_cmp3.bin.lzo of=./tag.bin bs=1K seek=544 count=80

/work/zeratul_t31/release/tools/make_tag/tag_generator/build_tag_t31_new.sh[POS=68,88][70%]16/06/21 - 17:36
```

3组相关的源文件

生成的tag.bin，384k之后进行lzo压缩处理，一组sensor所需的riscv+setting压缩之后的空间分配为80k。
根据需求可自行修改或者直接类比添加。

不同 sensor 主要的不同在 tag 中体现为以下 4 个文件：riscv 固件、效果文件、

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sensor init 文件、ae_table 文件。

如图所示，由于示例中只做了 3 款 sensor 的兼容，所以相关参数个数为 3 * 4 == 12，根据示例中的参数，参考 tag_generator.c 的代码，根据项目需求自行添加 sensor 的数量，相关的参数数量 == sensor num * 4。

确定好sensor 数量之后，将 tag_generator.c 的 MAX_NUM_SENSOR 改为一致。

```
/**      ...        **/
/**                 **/
/*********************/
/**                 **/
/**      END        **/
/**                 **/
/*********************///---------------------
/**                 **/
/**     RISCV FW    **///       128K risc-v fw
/**                 **/
/*********************///---------------------
/**                 **/
/**     MODE 0      **///       160K sensor setting
/**                 **/
/*********************///---------------------

#define MAX_NUM_SENSOR  3

#define RISCV_INTERVAL                  (288 * 1024)
#define SENSOR_INIT_INTERVAL            (2 * 1024)
#define SENSOR_INIT_ALL                 (16 * 1024)

#define SENSOR_AE_TABLE_INTERVAL        (2 * 1024)
#define SENSOR_AE_TABLE_SIZE            (2 * 1024)
#define SENSOR_AE_TABLE_ALL             (16 * 1024)

#define TAG_RISCV_FW_MAX_SIZE               (128 * 1024)

#define NUM_SENSOR_INIT_OFFSET          (352 * 1024)
#define NUM_SENSOR_TABLE_OFFSET         (NUM_SENSOR_INIT_OFFSET + SENSOR_INIT_ALL)
#define NUM_SENSOR_RISCV_OFFSET         (NUM_SENSOR_TABLE_OFFSET + SENSOR_AE_TABLE_ALL)
#define NUM_SENSOR_SETTING_OFFSET       (NUM_SENSOR_RISCV_OFFSET + TAG_RISCV_FW_MAX_SIZE)

#define MAX_TAG_ITER_SIZE (64 * 1024)
#define MAX_TAG_RISCV_FW_SIZE (128 * 1024)
#define MAX_TAG_SENSOR_SETTING_SIZE (160 * 1024)

#define MAX_TAG_SIZE (MAX_TAG_ITER_SIZE + MAX_TAG_RISCV_FW_SIZE + MAX_TAG_SENSOR_SETTING_SIZE + SENSOR_INIT_ALL + SENSOR_AE_TABLE_ALL + (RISCV_INTERVAL * MAX_NUM_SENSOR))
#define HALF_TAG_SIZE (MAX_TAG_ITER_SIZE / 2)
#define SEGMENT_SIZE (4 * 1024)
#define MAX_CMDLINE_LENGTH 1024
#define MAX_HEAD_SIZE 4 /* head size for each segment */
#define TAG_HEAD                        0xa108c309
#define TAG_END                         0x55aacc33

~/work/zeratul_t31/release/tools/make_tag/tag_generator/tag_generator_t31.c[POS=60,25][3%]16/06/21 - 20:55
```

3：sensor.sh 的使用 路径：userland/function/t23_multi_sensor

```
declare -x XMODIFIERS="@im=fcitx"
cywu@ingenic:~$
cywu@ingenic:~$
cywu@ingenic:~$ source sensor.sh
set num sensor config:
please enter max num of sensor : 3
set max mun sensor : 3
Supproted sensors:
  [1] gc2053
  [2] gc2063
```

先输入项目所需最大的sensor数，eg：需要3个则输入3.
后续依次按照指定顺序source sensor的型号

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```
[13] gc4653
[14] gc1054
[15] sc4236
[16] sc2335
[17] sc3335
[18] jxh62
[19] os02g10
[20] mis2006
[21] jxq03p
[22] sc301IoT
[23] gc4663

please enter num:1 sensor : 2
SENSOR = 2
RT_ENV_SENSOR_MULTI_MAX = 2
==============================================
RT_ENV_SENSOR_MULTI_0=gc4663
RT_ENV_SENSOR_MULTI_1=gc2063
RT_ENV_SENSOR_MULTI_MAX = 2
ksun@ksun-HP-EliteDesk-800-G1-USDT:~/zeratual_t41/release/userland/function/t41_multi_sensor$
```

注意：该脚本 source 的 sensor 型号的顺序与 common.c 的 sensor register 函数需要对应上。Eg：项目需要 gc4663 作为第一个 sensor，则此处第一个 sensor 选择 [23： gc4663]，与之对应需要在 comon.c 中 switch 到第一款 sensor num 之后将 gc4663 的驱动注册上。

```
common.c
" Press <F1> to display hel
common.c (/home/ksun/zeratu
|- variable
||   sensor_list
||   sensors
|
|- function
||   add_sensor_chose

37
38 //int init_sensor_one(void);
39 //extern int init_sensor_one(void);
40 int sensor_register_one(void);
41 int sensor_register_two(void);
42 int sensor_register_third(void);
43 struct list_head sensor_list;
44 //INIT_LIST_HEAD(&sensor_list);
45 struct add_sensor *sensors[8];
46
47 extern int use_num_sensor;
48
49 int add_sensor_chose(void)
50 {
51 //  printk("======= enter add_sensor_chose common  use_num_sensor = %d ======= \n", use_num
52     switch(use_num_sensor) {
53         case 0:
54             sensor_register_one();
55             break;
56
57         case 1:
58             sensor_register_two();
59             break;
60
61
62         default:
63             printk("====== swicth no match ====== \n");
64             break;
65
66     }
67 //  sensor_register_one();
```

此时这里对应 gc4663 驱动

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制作流程：

1：制作 tag：

1：source sensor.sh 根据需求选择对应 sensor 确定相关环境变量

2：source env_setup_devp.sh 选择需要的环境变量

3：在保证项目所需的各个 sensor 都有对应的 riscv、setting、init、ae_table 文件的前提下，在 tag_generator 目录下执行 build_tag_t23.sh

注意：制作 tag 时 tag_generator.c 内的 MAX_NUM_SENSOR 的宏需要对应修改。

2：制作驱动：

仔细阅读该文档 确认 sensor 驱动，isp 驱动、及相关文件 重新编译内核。

使用流程：

main app程序：更新 tag 分区 sensor 相关数据的主程序。

路径：userland/function/t23_multi_sensor/tag_updata_sensor_lzo_src

使用场景：

当设备更换一个新的 sensor 时，需要执行可执行程序 main，更新对应该 sensor 的相关文件，进行 tag 分区数据替换，执行完毕之后，flash tag 分区数据回写为当前 senor 所使用的相关配置，再次 reboot 即可正常出流。

注意：该功能由于涉及到部分闭源部分代码，为了后期维护方便，使用者需阅读文档了解操作流程之后，严格按照文档所制定的要求操作。

报错：

1.执行main程序时报错，需要打开i2c设备接口信息

```
[root@Zeratul:sdcard]# ./main
open i2c device failed:No such file or directory
i2c_init failed
INFO : check sensor index = 0  sensor.name = gc4663
INFO : tag_env_info --set HW use_num_sensor 0
Env Will Update
write_buf:S E N S
RISCV_OFFSET:544
DEBUG : output len = 65536
DEBUG : riscv head:RISC
DEBUG : write loop flash reisv ret = 65536
INFO :Firmware upgrade successful.
[root@Zeratul:sdcard]#
V:240409-T41ZG-af183f6-HP
adc:(4 -1 3300); EV:1951; is night:0; pwm duty:(0,0); while
mount: mounting /dev/mtdblock5 on /system failed: Invalid a
mount system failed
Setting up swapspace version 1, size = 16773120 bytes
UUID=bfef549e-19a9-4311-bb98-2f1c1f68abb2
```


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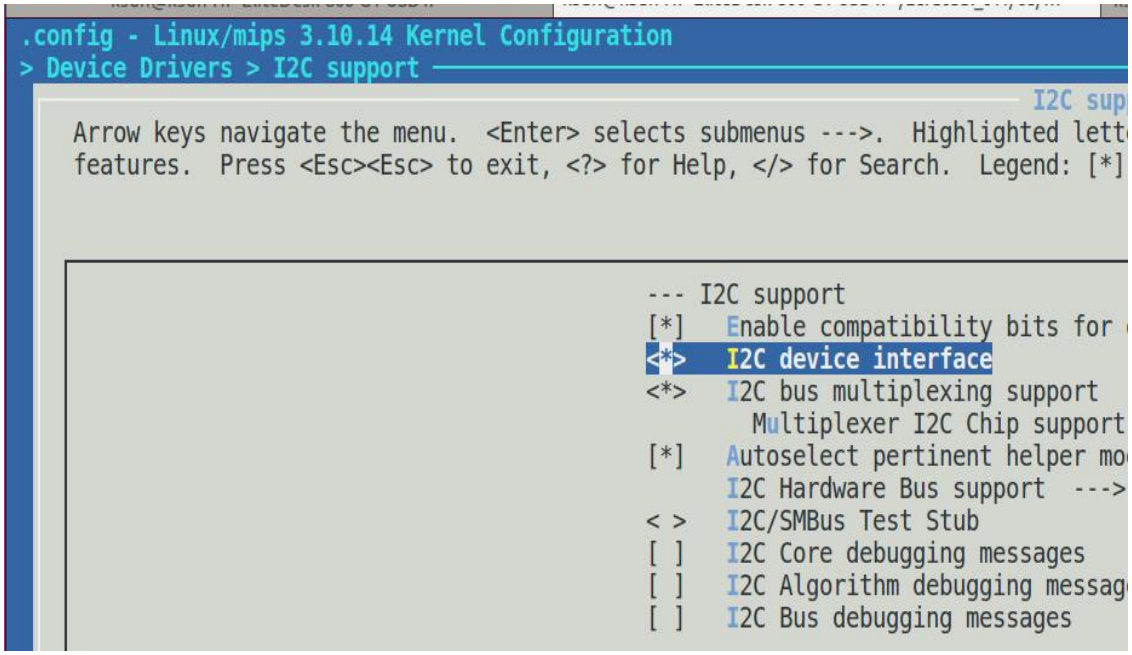

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